

turn water into steam, which drives an engine, and then turn the steam back into water again. But other processes, says the second law, go in one direction only. After you've mixed a jigger of gin into a glass of tonic water, there is no way you can separate the gin out again. All irreversible processes have one thing in common: the system moves toward a less ordered state. Entropy is the measure of the degree of disorder created.

Let's say you have a stack of pennies, all facing heads up, nice and orderly. Knock over the stack and some of the pennies will fall heads up and some tails up.

Now convert the pennies into stocks. If you sort through a thousand companies and stack, or rank, them according to their growth-in-earnings attractiveness, you'll have a Microsoft or its equivalent at the top and some low-growth dog at the bottom.

Suppose you bought them all and went to the South Seas for a year. Or five years. When you looked at your portfolio again, as with the pennies, you would see randomness, which would be more marked the longer you've been away. Entropy would be evident. Things just don't stay the same.

As time goes on, the top of the list will tend to drift down toward the middle. Growth stocks stumble. (It's a great temptation to buy a venerated name that's 15 percent cheaper than it was last week, but beware. It fell for a reason, probably an earnings disappointment, and one quarter's shortfall is usually followed by more. The Cockroach Theorem applies: "The chance of a company's having one and only one bad quarter is equivalent to the chance of your kitchen's having one and only one cockroach.") When I wrote my undergraduate thesis back in 1955, I chose two companies that were considered among the best in the world: Sears, Roebuck and Inland Steel. The steel industry was thought a high-growth business back then. And Sears dominated retail so completely that nobody could figure out how it could be stopped from taking over the country. Today Inland Steel and Sears, Roebuck may still be profitable, good companies, but if you were making a list of the twenty